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Data Science FINAL PROJECT REPORT

Project Title:

Wildfire Risk Prediction Using Climate and Environmental Data

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

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Abstract

Wildfires are a frequent and costly threat to the United States – a danger to citizens, to wildlife, to the landscape, and the wallets of taxpayers. This research project explores the idea that historic weather and environmental data can be utilized to classify a fire event as either a High Risk or Low Risk fire before it escalates beyond control using the principles of machine learning. This study uses the 1.88 million US Wildfires dataset published on Kaggle. It contains the number of fires that were opened as points with geo-location information from 1992 to 2015. The data underwent a careful cleaning and feature engineering process before being narrowed down to the final dataframe. It included the season and region information of a given case, as well as the engineered risk label – High and Low – that was derived from the size of the fire. Ridge Classifier was used as a linear baseline, while the data was evaluated using Random Forests, XGBoost, and Multi-Layer Perceptron algorithms. The training set used RandomUnderSampler due to High Risk fires being present in the data only fifteen percent of the time. All four models were trained on nine engineered features — including location, time, and cause — to predict a risk label, with High Risk being a positive class. The results were then visualized to show the accuracy of the sample data, with the most accurate model being selected for further processing. The fifth model, a proposed balanced bagging classifier, was trained using Optuna-tuned XGBoost trees as individual estimators with balancing applied within each bootstrap sample rather than across the entire dataset as done by RandomUnderSampler.

Across all five models, a balanced bagging classifier with an Optuna-tuned XGBoost base estimator demonstrated the best performance on the test set, achieving an accuracy of 73.46%, an F1-score of 0.4483, and a ROC-AUC of 0.8134. It outperformed the second-place model, the untuned XGBoost, with a margin of approximately one percent in terms of ROC-AUC, came significantly above the baseline ridge classifier and random forests, and surpassed the MLP classifier. Five-fold stratified cross-validation indicated that these results were consistent rather than an outlier-observation combination. ELI5 permutation importance analysis demonstrated that the absolute location of the fire, in longitude and latitude, was the most important feature in determining the risk of a fire, significantly outweighing the contribution of the fire causes, year, state, landowner category, the month, and season features.

Overall, the results demonstrate that, according to the analyzed dataset, the location of a fire is more important than the timing in determining its risk. This conclusion has clear implications for fire management, as it indicates that the resources should be allocated to studying particular areas rather than temporal phenomena in a specific region.

1. Introduction

1.1 Background

Wildfires are among the most insidious natural disasters in the United States, with a substantial number of large-scale wildfires already occurring in the past decade, and the risk of them happening only growing with the years. Large wildfires not only cause damage to the environment and the wildlife in them, but pose a significant risk to people and their dwellings. They also cause severe damage to the air quality of the affected regions and have begun to consume more and more of the emergency service budget with each following year. For these reasons, it is of paramount importance to ensure that the limited resources of emergency services are spent on only the most critical fires, as predicting the development of any fire is a resource-intensive process that cannot be afforded in all cases. In cases where the fire will most likely go out on its own, a large-scale response would only be a misallocation of resources.

With regards to machine learning, its application is obvious, as any emergency service will have copious amounts of data on fires that had occurred in the past, including fires of different scales and severity levels, as well as the ultimate outcomes for each. Therefore, if the patterns that determine whether a fire will escalate into a large-scale disaster or fizzle out on its own can be identified, they could be used to inform the decisions of emergency service personnel in future incidents. The use of machine learning would significantly increase the accuracy of the predictions while reducing the required input from firefighters, given that a majority of the necessary data would be collected for any fire by default. It could be used to determine which of the fires that have been reported will require emergency intervention, and which of them can be safely left to low-priority observation.

1.2 Motivation and Rationale for the Topic

This topic was chosen for three main reasons. First, wildfire risk classification sits at the intersection of environmental science and applied data science, allowing the full range of techniques taught in the Master of Data Science programme — data cleaning at scale, feature engineering, handling severe class imbalance, ensemble learning, hyperparameter optimisation, and model explainability — to be applied to a single, coherent, socially meaningful problem rather than a synthetic textbook dataset.

Second, the selected database, which is the 1.88 US Wildfires data set published on the crowdsourcing platform Kaggle, originally collected by the US Forest Service, seems to be peculiarly appropriate for the investigation carried out since it includes a substantial amount of data.. It is large enough (1.88 million individually recorded incidents spanning 24 years, from 1992 to 2015) to support robust training and testing of ensemble and neural-network models without needing to be artificially inflated, it is open and freely licensed (CC0), and it comes with a genuinely difficult and realistic class-imbalance problem — only around 15% of recorded fires fall into the higher fire-size classes that this project treats as "High Risk" — which makes the imbalance-handling techniques taught in the programme directly relevant rather than an academic exercise.

Third, wildfire prediction has real-world stakeholders. If a model that correctly identifies high-risk fires could be created, even if it were imperfect, it could plausibly help fire management agencies in triage decisions (e.g., resource allocation), insurance companies in risk assessment, or the general public in preparation and communication. The combination of technical depth, data, and applicability are why I chose this project over other options presented during the proposal phase.

Research Question

The project is guided by the following main research question: “How accurately can ensemble models and neural networks classify wildfire risk using historical climate and environmental data?”

1.3 Project Objectives

- OBJ-1: Compare ensemble models and a neural network to identify the best-performing model for wildfire risk prediction, using ROC-AUC and F1-score as the primary comparison metrics.
- OBJ-2: Identify the strongest environmental predictors of High Risk wildfire events using ELI5 permutation-importance explainability analysis.

1.4 Scope of the Study

The scope of this project was limited to binary risk classification (high risk vs low risk) rather than estimating specific fire sizes, and to the available tabular data (location, time, cause, ownership) rather than the weather and vegetation data not used for the last seven weeks for this project. The study only compared the results of five algorithms – Ridge_Classifier, Random_Forest, default XGBoost, MLP_Classifier, and the proposed Balanced_Bagging combination around the Optuna_XGBoost – without designing the actual predictive system for its continuous use.

2. Literature Review

The increase in the frequency and severity of wildfires in the western part of the US due to climate changes has led to the longer period of the fire season in recent years and the higher probability that the forest vegetation will be tinder-dry (Abatzoglou and Williams, 2016). These climate change effects contributing to the occurrence of potential wildfires are examples of phenomena that made possible to approach the research question from the perspective of prediction rather than just analyzing the past occurrences.

Jain et al. (2020) reviewed the broader use of machine learning across wildfire science, covering applications from occurrence prediction and burned-area estimation to spread modelling, and noted that tree-based ensemble methods have repeatedly outperformed simpler statistical approaches on structured, tabular fire-occurrence data of the kind used in this project. This is consistent with the design decision, in this project, to treat Random Forest and XGBoost as the strongest candidate baselines rather than relying on a purely linear model.

On the modelling side, Breiman (2001) introduced Random Forests as an ensemble of decorrelated decision trees that reduces the variance and overfitting risk of any single tree, which is why it was included here as the tree-based baseline. Chen and Guestrin (2016) introduced XGBoost, a scalable and regularised implementation of gradient boosting that has become a default choice for structured/tabular prediction tasks in both industry and academic competitions, motivating its use as both a standalone baseline and the base estimator inside the proposed final model in this project. For automated hyperparameter search, Akiba et al. (2019) proposed Optuna – a define-by-run optimisation framework, which unlike Grid Search or Random Search, allows narrowing the search area for each successive trial. This approach was chosen to tune the hyperparameters of the gradient boosting regressor due to the limited computational power of a Google Colab session. Another research direction related to the current one is addressing imbalanced classification problems. Indeed, the main technical challenge of the task at hand is recognising High Risk fires, which constitute a minority class. According to Lemaitre, Nogueira and Aridas (2017), who created the imbalanced-learn library, such scenarios can be addressed via RandomUnderSampler and BalancedBaggingClassifier algorithms, which were both used in the current project.

Finally, Korobov and Lopuhin's ELI5 library, which calculates permutation importance, was used to explain the features' impact on the model outcomes, thus addressing the second objective of the analysis. In particular, Sayad, Mousannif and Al Moatassime (2019) demonstrated that ELI5 can help identify important predictors for a gradient boosting model, presenting one of the more directly comparable studies, building a predictive wildfire model on a different regional dataset and machine learning pipeline, and reported similarly moderate absolute accuracy alongside stronger ranking-based metrics such as ROC-AUC — a pattern that also emerges in the results of this project, where accuracy is capped by the intrinsic difficulty and imbalance of the classification task, while ROC-AUC and F1-score provide a clearer picture of relative model quality. Goodfellow, Bengio and Courville's (2016) treatment of deep learning provided the theoretical grounding for including a Multi-Layer Perceptron as the neural network comparison point required by the research question, even though, as the results in Section 4 show, the MLP did not outperform the tree-based ensembles on this particular tabular dataset — a result that is itself consistent with the wider literature's observation that deep learning's advantages are most pronounced on unstructured data such as images and text, rather than structured tabular records.

3. Dataset and Methodology

3.1 Dataset Description

"The dataset "1.88 Million US Wildfires" (rtatman/188-million-us-wildfires), hosted on Kaggle was used for this project and downloaded with the kagglehub API. The data is a spatial database of wildfires compiled by Short (2017) for the US Forest Service Research Data Archive. The data spans 1992 to 2015 and was uploaded to kaggle as a SQLite database (FPA_FOD_20170508.sqlite). The data is not a CSV but rather a database so the entire Fires table was loaded into a pandas dataframe using an SQL query resulting in 1,880,465 rows and 39 columns prior to any cleaning.".

Each row contains information about one burned forest. The first column indicates the year and month when the fire was spotted. The second column provides the coordinates of the fire spot (the state too). The third column classifies fires by causes, the fourth – the owner category, and the fifth – FIRE_SIZE_CLASS, which is the basis for our binary target variable.

Data Cleaning

Rows that did not include any of the five columns vital for the target and core features, including FIRE_SIZE_CLASS, LATITUDE, LONGITUDE, STATE, and STAT_CAUSE_DESCR, were excluded from analysis because a target variable and location are essential elements of this task. In turn, the rows with missing information in the supporting feature, OWN_DESCR, were filled with the 'UNKNOWN' value, and duplicates were removed. For the current data set, the five key columns appeared to be complete, and there were no duplicates in the file. Therefore, the data set remained at the initial size of 1,880,465 rows, which also indicates that the source data were carefully prepared by their providers.

3.2 Feature Engineering

Several features were engineered from the raw data to generate additional predictive information beyond the original columns, namely:

- RISK_BINARY TARGET (High/Low): fire size class C-G (larger/moderately large fires) were assigned High Risk (1), while smaller/classes A-B fires were designated Low Risk (0). Within the cleansed dataset, this resulted in 1,606,295 Low Risk samples (85.4%) and 274,170 High Risk (14.6%), a ratio of ~6:1 which also affected the choice of the models below due to a strong class imbalance.
- SEASON / SEASON NAME: discovery day of the year (DISCOVERY_DOY) was binned into four seasons and label encoded as 0-3 for model training. With 607,092 Summer observations, 583,414 Spring, 419,295 Winter, and 270,664 Autumn, wildfires clearly peak during warmer months.
- REGION: states were grouped into 5 broad regions (West, Midwest, South, Northeast, or Other) per a predefined dictionary and then encoded using pd.factorize() as REGION_ENC. The South (950,358 records) and West (589,422) have a clear majority of fires in the dataset, which is unsurprising as the wildfire occurrences are strongly associated with those regions.
- Label encoding: REGION, OWNER_DESCR, STATE, STAT_CAUSE_DESCR – all categorical fields with string values – were converted to numerical domain using LabelEncoder from scikit-learn, resulting in REGION_ENC, OWNER_ENC, STATE_ENC, CAUSE_ENC variables. The reason for this approach was the requirement of the model algorithm to accept only numerical features.

3.3 Exploratory Data Analysis

Five static plots and one interactive geospatial visualization were constructed to explore the data prior to attempting any sort of modelling: a bar plot visualizing the Low Risk / High Risk class distribution (confirming the 85.4% / 14.6% discovered above), a year plot, again confirming the visualized trend from the descriptive statistics, a season plot, also confirming the descriptive finding of concentration on summer/spring, a cross-tabulation of region and risk class, visualizing the difference in risk class distribution between geographical areas, and, finally, an interactive Folium heatmap, which was used to identify the regional concentration of High Risk fires, which would later be formalized in the ELI5 analysis. After all plots were finalized, they were saved as high-resolution PNG images and the Folium heatmap saved as an HTML file to the project's Google Drive output directory.

Nine features have been selected for modelling: FIRE_YEAR, MONTH, SEASON, CAUSE_ENC, LATITUDE, LONGITUDE, REGION_ENC, OWNER_ENC, STATE_ENC. The tenth feature, FIRE_SIZE_LOG, a transformed value of the raw size of fires, has been intentionally removed. The reason is that the feature FIRE_SIZE_CLASS, which indicates the size class of a fire, is closely related to the raw value, not the transformed one. Thus, using FIRE_SIZE_LOG would lead to data leakage, making the model's performance artificially high. The data was split, seventy percent for training, and thirty percent for testing (1,316,325 and 564,140 rows). Stratified sampling was used so that the proportion of risk classes, approximately eighty-five to fifteen, would be similar in both the train and test sets. The standard scaler was applied to the train set and then to the test set, not the other way around, as is the standard practice.

Handling Class Imbalance

Because High Risk fires comprise only 15% of the data, training a model directly on the raw training set would result in a classifier that always predicts “Low Risk” for most cases and would achieve decent accuracy on the test set. Two approaches have been taken to evaluate this issue. For the four baseline models (Ridge, Random Forest, default XGBoost, and MLP), a RandomUnderSampler has been applied to the training data only, resulting in a sample that contains 191,919 Low Risk and 191,919 High Risk fires, 383,838 records in total (the original train set had 1,124,406 Low Risk and 191,919 High Risk fires — please re-verify this split against your notebook's .value_counts() output). In this case, only the training set has been resampled, not the test one – which means that the performance of such a classifier has been evaluated on the real, imbalanced distribution of records. For the proposed final model, a different approach has been taken. Instead of under-sampling the training set at the beginning of training, a BalancedBaggingClassifier has been used. It trains a collection of classifiers (estimators) on different bootstrap samples of the original training data. Each estimator in the ensemble was trained on an equal mixture of the data (High and Low risk fires were split in half) and rebalanced internally within each bag before fitting the base estimator. The model was trained on the original imbalanced 1316325-row training set, with balancing inside each bag, but not for the whole set, as was the case for the four simpler models above.

3.4 Models Compared

Five modelling approaches were trained and evaluated, in the following order:

- Model 1 — Ridge Classifier (baseline): a linear classifier with L2 regularisation (alpha = 1.0), included as the simple baseline that every more complex model is expected to beat.
- Model 2 — Random Forest: an ensemble of 200 decision trees, chosen for its established robustness to noisy, structured tabular data and its resistance to overfitting relative to a single tree.
- Model 3 — XGBoost (default parameters): a gradient-boosted tree ensemble (300 boosting rounds, maximum tree depth of 8, learning rate 0.05, with 80% row and column sub-sampling per tree), used first with reasonable hand-picked defaults as a baseline.
- Model 4 — Multi-Layer Perceptron (MLP): a two-hidden-layer neural network (128 neurons then 64 neurons) with early stopping enabled, representing the deep-learning comparison point required by the research question.
- Model 5 —Balanced Bagging + XGBoost (proposed final model): a BalancedBaggingClassifier implementing the Optuna-tuned XGBoost as the base_estimator, fit to the initial unbalanced training set with internal balancing of the bagged samples as described in the Handling Class Imbalance subsection of section 3.3. This model constitutes the proposed answer to the posed research question by this project, and as such it is the one that will be saved for later use.

3.5 Hyperparameter Tuning with Optuna

Rather than manually tweaking the hyperparameters or using the exhaustive Grid Search, for this project, an automated hyperparameter optimisation called Optuna was used. Specifically, it searched through 9 hyperparameters of the XGBoost classifier, namely, number of estimators, maximum tree depth, learning rate, subsampling rates of rows and columns, minimum weight of child, minimum loss reduction per leaf gamma, and regularisation L1 and L2 parameters. These hyperparameters were tuned using the Optuna framework by performing 30 trials on 3-fold stratified cross-validated balanced training samples (rather than all the training data, which would give an unrealistically high ROC-AUC of around 1) in order to find the set of hyperparameters that maximised the mean ROC-AUC across folds.

3.6 Explainability with ELI5

To satisfy the second objective of the project, namely determining the top environmental predictors of High Risk wildfires, the permutation-importance procedure was used on the tuned down XGBoost model. Specifically, the procedure uses the hold-out sample (in this case, a randomly chosen 30,000-row subset of the test set), where it measures the decrease in ROC-AUC when a feature's values are permuted: the bigger the drop, the more informative the feature is for prediction. To make the procedure less computationally intensive, only 5 permutations were made for each feature to estimate the importance score and standard deviation..

Evaluation Strategy

Each model was evaluated on the same untouched class-imbalanced test set using three different metrics, Accuracy, F1-score (for the minority High Risk class, which is more informative given the context of the problem), and ROC-AUC (an ability of the model to rank positives higher than negatives, which is a valuable metric given the importance of correctly classifying High Risk). Accuracy on an imbalanced data set is a misleading metric, hence why we focused on evaluating models on a ROC-AUC and F1-score, which were mentioned explicitly in the research question. For the final comparison between the models, we used 5-fold stratified cross validation to assess if the ROC AUC of each model was not overestimated on the particular 70/30 split of the data, which is a common issue when using the hold-out method. Lastly, we compared each model's performance on train and test sets to ensure that the models were neither underfit nor overfit the data.

4. Results

4.1 Baseline Model Comparison (Before Tuning)

The four baseline models were first evaluated on the untouched test set, all trained on the RandomUnderSampler-balanced training data described in Section 3.3. Table 1 summarises the results.

Model	Accuracy	F1-Score	ROC-AUC
Ridge (Baseline)	0.5639	0.2823	0.6005
Random Forest	0.7169	0.4279	0.7943
XGBoost (Default)	0.7109	0.4276	0.7972

MLP (Neural Network)	0.6854	0.4036	0.7718

Table 1. Baseline model comparison on the test set, before Optuna tuning.

By ROC-AUC, default XGBoost was the strongest of the four baselines (0.7972), narrowly ahead of Random Forest (0.7943), with the MLP some way behind (0.7718) and the linear Ridge classifier clearly the weakest (0.6005) — an expected result given that Ridge cannot capture the non-linear relationships between geography, season, and fire risk that the tree-based models can. Because XGBoost was the strongest baseline family, it was selected as the candidate for Optuna hyperparameter tuning and as the base estimator for the proposed final model.

4.2 Optuna Hyperparameter Tuning

Thirty Optuna trials, each evaluated with 3-fold cross-validation on the balanced training data, produced a best mean cross-validated ROC-AUC of 0.8008. The winning hyperparameter combination is shown in Table 2.

Hyperparameter	Tuned Value
n_estimators	421
max_depth	10
learning_rate	0.1168
subsample	0.8346
colsample_bytree	0.9326
min_child_weight	10
gamma	0.4930
reg_alpha	1.94e-06
reg_lambda	0.000213

Table 2. Optuna's best XGBoost hyperparameters after 30 trials.

The accuracy of XGBoost, rebuilt using the optimal parameters found and evaluated on the test set, was 0.7173, F1-score was 0.4364, while the ROC-AUC score was 0.8064. Comparing these results to the default XGBoost model with no parameter tuning, the improved accuracy is only 0.0064, while the AUC score increased by 0.0092. Thus, the parameter tuning did have an effect, although it can be considered a relatively small one.

4.3 Proposed Final Model and Overall Comparison

The proposed final model, i.e., the BalancedBaggingClassifier of the 10 bags, each of which is the Optuna-optimized XGBoost classifier trained on the imbalanced train set, achieved the accuracy of 0.7346, F1-score of 0.4483, and ROC-AUC of 0.8134 on the test set. Table 3 below summarizes the results of all the models, thus addressing the first project's research question.

Model	Accuracy	F1-Score	ROC-AUC
Ridge (Baseline)	0.5639	0.2823	0.6005
Random Forest	0.7169	0.4279	0.7943
XGBoost (Tuned)	0.7173	0.4364	0.8064
MLP (Neural Network)	0.6854	0.4036	0.7718
Balanced Bagging + XGBoost (Proposed)	0.7346	0.4483	0.8134

Table 3. Final comparison across all five models on the test set.

The proposed Balanced Bagging + XGBoost model was the best-performing model overall by both ROC-AUC and F1-score, outperforming the strongest plain baseline — tuned XGBoost — by 0.0070 ROC-AUC and 0.0119 F1-score. This confirms the value of the internal, per-bag balancing strategy over the simpler single-pass under-sampling used for the four baseline models, and directly answers Objective 1: among all models compared, the Balanced Bagging ensemble around a tuned XGBoost base learner is the best-performing model for wildfire risk classification on this dataset.

4.4 Cross-Validation Stability

To confirm that these results were not an artefact of the particular 70/30 train-test split used, 5-fold stratified cross-validation was run on the balanced training data for the four models compatible with the standard cross_val_score workflow (the proposed Balanced Bagging model, which uses a non-standard internal balancing step, was evaluated separately via the train/test split above rather than through this cross-validation loop). The results, in Table 4, are similar to the single-split test results, and the standard deviations are small across all, which suggests consistency and reliability rather than an accidental split. To further verify that these results are not due to a chance split, another split was performed.

Model	Mean ROC-AUC	Std. Dev.
Ridge (Baseline)	0.6008	0.0014
Random Forest	0.7902	0.0010
XGBoost (Tuned)	0.8034	0.0011
MLP (Neural Network)	0.7676	0.0012

Table 4. Five-fold stratified cross-validation results (ROC-AUC, mean ± standard deviation).

4.5 Overfitting and Underfitting Analysis

Comparing each model's accuracy on the data it was trained on against its accuracy on the untouched test set gives a simple check for overfitting (a large gap, where the model memorises training data but does not generalise) or underfitting (both scores are low, meaning the model is too simple to capture the pattern at all). Table 5 summarises this comparison.

Model	Train Accuracy	Test Accuracy	Gap	Verdict
Ridge (Baseline)	0.5731	0.5639	0.0092	Underfitting
Random Forest	0.9969	0.7169	0.2800	Overfitting
XGBoost (Tuned)	0.7839	0.7173	0.0666	Generalises well
MLP (Neural Network)	0.7096	0.6854	0.0242	Underfitting
Balanced Bagging + XGBoost	0.7485	0.7346	0.0138	Underfitting

Table 5. Train-versus-test accuracy comparison across all five models.

The Random Forest method demonstrates the biggest training/test gap, which is unsurprising given its unrealistically high train accuracy of 0.9969, suggesting that with 200 unconstrained trees it has fallen victim to overfitting the training set. Ridge and the MLP have only slightly bigger gaps between their absolute accuracy scores on the two sets, but perform only moderately well on both, suggesting that they are insufficiently expressive models for this task (Ridge is linear, after all) or their regularization erroneously constrains their capacity to learn the wildfire risk patterns. XGBoost has the smallest gap combined with decent absolute accuracy on both sets, which speaks for its overall performance being the closest to ideal. The Balanced Bagging approach has a similarly small gap and highest test accuracy among all models, and its overall performance is corroborated by having the highest ROC-AUC and F1-score in Table 3; however, its train accuracy is reasonably low, with the automated note in the notebook calling it "underfitting", not "balanced".

4.6 Explainability: Top Environmental Predictors (Objective 2)

ELI5 permutation importance was applied to the tuned XGBoost model to identify which of the nine input features it relied on most heavily. Table 6 presents the resulting importance scores, sorted from most to least influential.

Feature	Importance (ROC-AUC drop)	Std. Dev.
LONGITUDE	0.1838	0.0048
LATITUDE	0.1272	0.0036
CAUSE_ENC (fire cause)	0.0372	0.0014
FIRE_YEAR	0.0371	0.0012
STATE_ENC	0.0349	0.0010
OWNER_ENC (landowner)	0.0296	0.0014
MONTH	0.0224	0.0012
REGION_ENC	0.0036	0.0003
SEASON	0.0023	0.0003

Table 6. ELI5 permutation importance of the tuned XGBoost model, sorted by importance.

The two location features, LONGITUDE (0.1838) and LATITUDE (0.1272), are by a wide margin the strongest predictors of wildfire risk — together they account for far more of the model's predictive power than any of the other seven features combined. This is consistent with the geographic concentration of High Risk fires already visible in the EDA (Section 3.3), and directly answers Objective 2: precise geographic location, more than the recorded cause, the year, the state, the landowner, the month, or even the season, is what the tuned model relies on most when distinguishing High Risk from Low Risk fires. Interestingly, the two most explicitly "climate-flavoured" engineered features — REGION_ENC and SEASON — were the two least important of all nine, suggesting that the fine-grained geographic signal captured by raw latitude and longitude already subsumes most of the information that the coarser, hand-engineered REGION and SEASON groupings were designed to capture.

5. Discussion

The findings reported in Section 4 address both of the project's objectives. With regard to Objective 1, it is found that the proposed Balanced Bagging + Optuna-tuned XGBoost was the best-performing method across all three metrics, with the improvements over the baseline of the standard XGBoost indicating the value of incorporating bagging with internal balancing compared to using undersampling at the beginning of the pipeline, for this particular imbalanced dataset. However, the magnitude of this performance gain was relatively modest (about 0.007 ROC-AUC points), and the overall predictive performance, while reasonable, was still limited, with accuracy in all models peaking at 73%. This limitation was determined to be mostly inherent to the problem, as there are many factors that influence the fire size category beyond the ones encoded in this particular dataset, namely, the local weather, the state of the fuel supply, wind, and the response rate of the firefighting teams, which were not available as features. Another finding pertinent to the interpretation of the results was that across all models, the precision for the High Risk class was substantially lower than recall, meaning that fewer fires classified as High Risk were indeed high risk (the figure for the proposed method was 0.3216, or roughly one third), which could be a serious limitation depending on the context. Specifically, in a triage context, the value of the likelihood of a fire being high risk would have to be balanced against the opportunity cost of allocating resources to fires that turned out not to be, and the operating threshold for High Risk would need to be adjusted depending on the circumstances.

Finally, the finding that the MLP network failed to compete with the tree-based methods was in line with the general trends reported in the literature, namely, that the deep learning methods tend to require much more data to produce meaningful results, and typically excel in unstructured data such as images or audio, while tree-based models retain advantages in tabular data. This pattern was confirmed again by the present study. With regard to Objective 2, the findings in Section 4.6 demonstrated one practical application of the explainable AI methods, specifically, the fact that the influence of the geographic grouping feature far outweighed the influence of the temporal features of the fire (month, season) suggested that the resource allocation decisions should consider prioritizing the areas with historically higher fire severity rather than seasonal patterns. It is noted, however, that this finding was based on data up to 2015, and the seasonal trends could have changed since due to climate change, which is not accounted for in this project.

6. Conclusion and Future Work

This project aimed to evaluate the accuracy of the ensemble and neural network models for the task of classifying wildfires based on historical weather and environmental data and identify environmental factors associated with them. For this purpose, the US Wildfires dataset containing 1.88 million instances was used. The Ridge, Random Forest, XGBoost, MLP, and proposed Balanced Bagging of Optuna-tuned XGBoost Model were trained and compared. The proposed model demonstrated the best performance with an ROC-AUC of 0.8134 and an F1-score of 0.4483. Due to the cross-validation procedure, it was possible to ensure that the achieved accuracy was not due to overfitting. It was found that the data on the geographical location (latitude and longitude) was the most informative one for the prediction of High Risk. The value of the temporal features (Month, Season, Year) was significantly lower. Therefore, these findings might be considered for further research.

Several recommendations for future research were proposed. Firstly, it was suggested to include the set of time-dependent external variables (temperature, wind speed), humidity, and drought indices, which are more related to the fire spread, rather than historical features used in the current study. Secondly, there was an idea to transform the binary classification task (High/Low) into a multi-class or continuous variable to predict the size of a fire with greater precision. Finally, the authors suggest testing the proposed model on the more recent data to check whether the identified geographical patterns hold true for the present time since the data used in the study was available until 2015.

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